Where Teachers Teach: Position as the Signal Indicator in On-Policy Distillation

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Abstract

On-policy distillation (OPD) trains a student to match a teacher on the student's own samples by minimizing reverse KL token-by-token. The default loss weights every generated token equally; we ask: *among all generated tokens, which ones actually carry the distillation signal?* Holding the per-response budget fixed at K=100 tokens, we benchmark three families of selectors against the full-sequence loss on MATH-500: top-K by per-token KL, three entropy-based variants (top-K student entropy, top-K teacher entropy, learned format-mask), and a contiguous prefix of the first N positions. Top-KL underperforms full-seq by 3.75 pp despite covering 93% of the per-token KL mass between the *initial* student and teacher; entropy variants all land within $\pm$1 pp of full-seq; only *prefix-K exceeds full-seq*, by 3.5–4.3 pp on math. Position is not a coverage proxy: middle-100 and last-100 fall *below* the no-distill baseline.

Prefix-K improves over full-sequence in 7 of 9 cells we test across model families (Qwen, Gemma, Llama), training methods (LoRA, FullFT), and tasks (math, code, BFCL function calling); the two cells where prefix is not the winner are FullFT-math (loses by 1.5 pp) and Llama BFCL (where neither full-seq nor prefix-K beats the teacher). It cuts wall-clock by $\sim$10$\times$ and peak memory by $\sim$4$\times$, requires a one-line code change, and exhibits a single elbow at $N \approx$ 50–150. On BFCL function calling, prefix-K students exceed their teachers in two of three pairs (by 5.9–7.3 pp). We give two interpretations consistent with the data: a cascade by which prefix supervision auto-aligns the tail (verified by tail-KL drop and a prefix-swap experiment), and mode-seeking reverse-KL on a contiguous prefix concentrating the student on a high-conviction trajectory mode.

1 Introduction

On-policy distillation [Agarwal et al., 2024, Gu et al., 2024, DeepSeek-AI, 2025] trains a student on its own samples scored by a stronger teacher, and is now a workhorse for transferring reasoning and structured-output capabilities into smaller models. The standard objective is a *token-wise* reverse KL averaged over all positions of each on-policy trajectory. The default weighting—one per token—bakes in the assumption that every position contributes equally to the supervision signal.

This assumption is empirically fragile. Across our experiments, full-sequence OPD often degrades during training: with three seeds for Qwen2.5-Math-1.5B→Qwen3-1.7B on MATH-500, mean accuracy is 56.78 ± 6.1 pp and one seed falls *below* the no-distill baseline; for Gemma-2-2B→Gemma-3-4B math, full-seq sits at 11.70% against a 13.45% no-distill baseline; for Llama-3.2-1B→Llama-3.1-8B function calling, full-sequence distillation ends 23.3 pp *below* the raw student (raw 55.3%, post-distill 32.0%); for Qwen coding, HumanEval degrades from 39% to 32% over training. These are not implementation bugs—they are symptoms of a structural problem with the assumption that all token positions carry equal supervision value.

The question. Rather than patch the loss with a hand-crafted reweighting, we ask: *which tokens carry the distillation signal?* We split the question into a head-to-head comparison among three plausible answers, holding the per-response budget fixed at K=100 tokens:

- **KL**: select the top-K tokens with highest per-token KL—a natural divergence-based choice.
- **Entropy**: top-K by student or teacher entropy, or a learned *format-mask* that drops low-entropy format tokens.

- **Position**: the first K tokens of each response (prefix-K).

A good indicator should at least *exceed* full-sequence training; anything below full-seq is a strict regression—you would be better off using every token.

Findings. Only position passes that bar. On MATH-500 with the canonical Qwen pair (Table 1), top-KL-100 underperforms full-seq by 3.75 pp despite covering 93% of total per-token KL mass; the three entropy variants land within ± 1 pp of full-seq; prefix-K exceeds full-seq by 3.5–4.3 pp at $N \in \{50, 100, 150, 200\}$. Negative controls (middle-100, last-100) fall *below* no-distill, ruling out the hypothesis that the gain is just a coverage effect. The prefix wins despite covering only 46% of per-token KL mass.

Generality. Prefix-K matches or improves over full-sequence across model families (Qwen, Gemma, Llama—in two of three, full-sequence *collapses*), tasks (math, code, BFCL), and methods (LoRA, full fine-tuning); the one cell where prefix loses is FullFT-math (Table 4). It is also $\sim 10\times$ faster wall-clock, $\sim 4\times$ smaller in peak GPU memory, requires a one-line code change, and shows a single performance elbow at $N \approx 50$–150 across tasks. On BFCL function calling, students trained with prefix-K *exceed* their teachers in two of three pairs (by 5.9–7.3 pp).

Contributions. (1) The first head-to-head comparison of KL-, entropy-, and position-based selectors in OPD at fixed budget, revealing that only position exceeds full-sequence. (2) Negative controls (middle/last) showing that late-token training is *actively harmful*—worse than no distillation. (3) Cross-family / cross-task / cross-method validation, including stability and efficiency. (4) A cascade-plus-mode-seeking explanation backed by a tail-KL drop without tail training, a prefix-swap experiment, and token-level statistics from `signal_analysis/range_summary.json`.

2 Setup

On-policy distillation. The student p_θ samples a response $y \sim p_\theta(\cdot \mid x)$; the teacher q_ϕ scores every position. The default loss is a sum of per-position reverse KL over the full response of length T:

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{y \sim p_\theta(\cdot|x)} \left[\sum_{t=1}^{T} D_{\text{KL}}(p_\theta(\cdot \mid x, y_{<t}) \,\|\, q_\phi(\cdot \mid x, y_{<t})) \right]. \tag{1}$$

A token *selector* replaces the uniform sum by a sum over a subset $\mathcal{S}(x, y) \subseteq \{1, \ldots, T\}$ of size K: $\mathcal{L}_{\mathcal{S}} = \mathbb{E}\left[\sum_{t \in \mathcal{S}} D_{\text{KL}}(\cdot)\right]$. We compare:

- **Top-KL-K**: $\mathcal{S}$ is the top-K positions by $D_{\text{KL}}(p_\theta \| q_\phi)$ on the sampled trajectory.
- **Top-Ent-Student-K**, **Top-Ent-Teacher-K**: top-K by student / teacher entropy.
- **Format-mask**: drop low-entropy format tokens above a learned threshold (heuristic for "style" tokens).
- **Random-K**: a K-uniform-random subset (control).
- **Prefix-K**: $\mathcal{S} = \{1, \ldots, K\}$, the first K generated tokens.
- **Middle-K**, **Last-K**: contiguous middle / final K positions (negative controls).

Notation. K denotes the per-response selector budget; for prefix-K, $K = N$ (the prefix length). Outside of selector tables we use N to emphasize the contiguous-prefix view, and K where contrast with non-prefix selectors is the point. "Pos-200tok" is a prefix selected by token-count rather than position-index; in our setting they coincide for tokenizer-aligned prompts and we use the labels interchangeably.

Models. The canonical pair is Qwen2.5-Math-1.5B [Qwen Team, 2024] as student, Qwen3-1.7B [Qwen Team, 2025] as teacher; we also use Gemma-2-2B→Gemma-3-4B and Llama-3.2-1B→Llama-3.1-8B for cross-family results.

Training. LoRA [Hu et al., 2022] with $r = 32$, $\alpha = 64$ on $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$_proj; learning rate 5×10^{-5}; 200 steps; 3200 problems; $n = 1$ on-policy sample per problem; effective batch size 16 (**n1bs16**). Generation temperature 0.7. We also report full fine-tuning (FullFT) results. For position-restricted runs, we additionally clamp `max_new_tokens` to N, which is where most of the wall-clock savings come from.

Table 1: Selector head-to-head, MATH-500 avg@4, n1bs16 LoRA, K=100. Only the contiguous prefix *exceeds* full-sequence. Negative controls (middle/last) fall below the no-distill baseline. KL coverage = fraction of total per-token reverse KL between the *initial* student and teacher selected by the indicator on held-out trajectories; not invariant under training.

Selector	**avg@4**	**Δ vs full-seq**	**KL covered**
No-distill baseline	50.95	−11.40	—
Middle-100 (below baseline)	47.80	−14.55	∼30%
Last-100 (below baseline)	50.35	−12.00	∼15%
Top-KL-100	58.60	−3.75	93.2%
Top-Ent-Student-100	61.35	−1.00	67.0%
Format-mask	62.05	−0.30	—
Top-Ent-Teacher-100	62.20	−0.15	50.5%
Full-sequence	62.35	0.00	100.0%
Random-100	63.05	+0.70	21.1%
Pos-100	65.85	+3.50	45.6%
Pos-200tok	66.05	+3.70	—
Pos-50 / **Pos-150**	**66.65**	**+4.30**	—

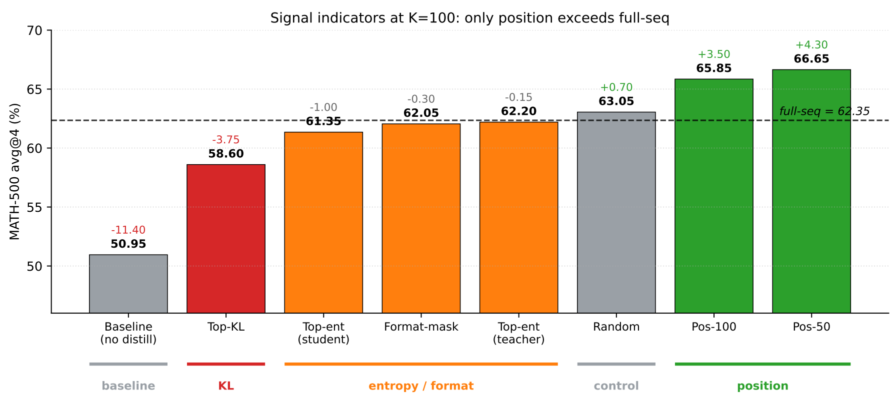

Figure 1: Signal indicators at fixed K=100. Bars show MATH-500 avg@4 for each selector; the dashed line is full-sequence. Top-KL underperforms; entropy variants tie; only prefix exceeds. Negative controls fall below baseline.

Evaluation. MATH-500 [Lightman et al., 2023, Hendrycks et al., 2021] reported as avg@4 / maj@4 / pass@4 with $n = 4$ samples at $T = 0.7$; HumanEval / MBPP [Chen et al., 2021, Austin et al., 2021] via `evalplus`; BFCL function-calling full-accuracy. The undistilled student baselines are 50.95% (Qwen math), 13.45% (Gemma math), and 15.20% (Llama math). The full-sequence number for the canonical Qwen pair is 62.35%.

3 Part 1: What is a Good Signal Indicator?

We hold $K = 100$ fixed and ask which selector exceeds the full-sequence baseline. Table 1 is the headline result; Figure 1 visualizes it.

3.1 KL is the Wrong Indicator

The most natural indicator—rank tokens by their reverse KL—underperforms full-sequence by 3.75 pp, despite covering more than 90% of total per-token KL mass. Token classification (Table 2) explains why: the highest-KL tokens are dominated by LaTeX format and structural style, not reasoning. The category `math_latex` (e.g., `\(`, `\[`, `\\`) has mean KL 3.99—more than 10× that of digits and operators. Training the student on these positions teaches it how the teacher *types*, not how it *thinks*.

Table 2: **Token classification at high-KL positions.** Highest-KL = format / style disagreement, not reasoning quality.

Category	Example tokens	Mean KL
math_latex (format)	`\(`, `\[`, `\\`	3.99
planning	"To", "First", "Therefore"	1.75
structural	`**`, `:`, newlines	0.89
math_operator	$=, +, -$	0.38
math_number	0–9	0.28

Takeaway. High KL is a divergence metric; it is not, by itself, a teaching signal.

3.2 Entropy: Necessary, Not Sufficient

If high-KL fails because LaTeX format dominates, then entropy—which downweights low-entropy format tokens—should win. It does not. The three entropy-based selectors (top-K student entropy, top-K teacher entropy, format-mask) all land within ± 1 pp of full-sequence (Table 1). They strip out the worst format tokens that hurt top-KL but do not exceed full-seq. Even random-K matches full-seq—suggesting full-sequence's number is a *floor* for any non-pathological selector that avoids format-dominated tokens, not a ceiling.[1]

Why entropy ties. Entropy ranks tokens by local uncertainty, which loosely correlates with where the teacher has something to teach. But a high-entropy token *late* in a trajectory sits on a path that has already drifted; aligning it does not fix the answer. A high-entropy token *early* sets the trajectory; aligning it can fix everything downstream. Top-entropy weights both equally, washing out to full-sequence behavior.

3.3 Position Uniquely Exceeds Full-Sequence

Prefix-K is the simplest possible selector: take the first K generated tokens. It exceeds full-sequence by 3.5–4.3 pp at $K \in \{50, 100, 150, 200\}$ and is the only selector with this property in our experiments.

Position is not just a high-entropy region. If position were *only* entropy-correlated, hard top-entropy-K should match prefix-K. The 4.5 pp gap (61.35 vs. 65.85) shows it does not. Inside the head (Appendix B): the first 50 tokens are 2.4× enriched in the top global surprise quartile but still contain 13.1% bottom-quartile tokens; even head-500 covers only 31.8% of the top quartile. Position is correlated with entropy, not equivalent to it.

Why does the contiguous prefix dominate? We see two effects, taken up in detail in Section 5:

1. **Causal coverage.** Aligning the planning prefix re-orients the entire trajectory; aligning scattered late tokens does not (cascade evidence: §5.1).
2. **Distribution-shape coupling.** A contiguous prefix matches the autoregressive structure of the model: every gradient step aligns the conditional $p_\theta(y_t \mid y_{<t})$ at the same context the model actually decodes from at inference. Mode-seeking reverse-KL on this prefix concentrates the student on a single high-conviction trajectory mode (§5.2).

Bridge to Part 2. The Part 1 result is established on a single setting (Qwen-math, n1bs16 LoRA, K=100). Before unpacking the mechanism, we test how broadly it survives: across model families, tasks, training methods, seeds, and choices of N.

4 Part 2: Robustness of the Part 1 Claim

Part 1's head-to-head was on one pair, one task, and one seed configuration. This section tests whether the prefix-K wins survive across model families, tasks, training methods, seeds, and choices of N, and whether they come with practical efficiency benefits worth caring about.

[1] We deploy entropy and confidence as fixed-budget hard selectors here; the cited works in §6 use them as soft reweightings of the full-sequence loss. The two are not interchangeable, and our claim is the narrower one (§6, "Scope of comparison").

Table 3: MATH-500 (Qwen2.5-Math-1.5B→Qwen3-1.7B, n1bs16 LoRA). Prefix-K improves over full-seq by 3.5–4.3 pp on avg@4 and 4.3–7.4 pp on pass@4. Best step among checkpoints {50, 100, 150, 200}.

Method	**Best step**	**avg@4**	**maj@4**	**pass@4**
No-distill	—	50.95	61.20	72.80
Top-KL-100	50	58.60	65.80	74.40
Top-Ent-Student-100	100	61.35	67.20	73.20
Format-mask	50	62.05	67.20	76.20
Top-Ent-Teacher-100	150	62.20	67.80	75.80
Full-sequence	150	62.35	69.40	74.60
Random-100	150	63.05	69.20	76.80
Pos-100	200	65.85	70.80	79.80
Pos-200tok	50	66.05	71.20	81.00
Pos-50	150	**66.65**	71.00	**81.00**
Pos-150	100	**66.65**	67.00	**81.00**

4.1 Main Results

Table 3 reports MATH-500 results for the canonical pair across selectors and position settings.

Coding (HumanEval, LoRA). On the coding task, LoRA full-sequence *degrades* from 40.2 (step 50) to 26.8 (step 400)—a 13.4 pp drop over training. Prefix-K is stable: pos-50 reaches 42.1 (HE) / 36.6 (HE+) at step 350; pos-100 reaches 42.1 / 36.0 at step 150. Full results in Appendix A.

Negative controls. Middle-100 reaches only 47.80 pp—3.15 pp *below* the no-distill baseline—and last-100 reaches 50.35, also below baseline (Table 1). The contrast with prefix-100 (65.85) cannot be explained by training-budget or coverage arguments alone: the same number of tokens placed in different parts of the response yield differences of >15 pp. Training on late tokens is not just less useful—it is *actively harmful*.

4.2 Generality: Cross-Family, Cross-Task, Cross-Method

Table 4 sweeps families, tasks, and methods. Three observations: (i) prefix-K wins on math for every family, in two cases where full-sequence *collapses below baseline*; (ii) prefix-K matches or wins on coding and BFCL function calling, with one BFCL pair (Llama, Table 7) where prefix falls 4.7 pp below the teacher; (iii) the one cell where prefix *loses* is FullFT-math (56.75 vs. 58.20, −1.45 pp). CoT ("thinking") mode shows the same pattern as plain decoding: Qwen3-1.7B→Qwen3-4B with `enable_thinking=True`, prefix-100 reaches 66.80 vs. 64.65 for full-seq.

Where prefix doesn't win. We do not claim prefix-K is universally better. Two cells deserve named explanations. *FullFT-math.* Full fine-tuning at the same step count uses ∼10× smaller learning rate (5×10^{-6} vs. LoRA's 5×10^{-5}); we conjecture the smaller per-token effective update tolerates noisy late-token gradients better, narrowing the prefix advantage. *Llama BFCL.* The Llama 1B→8B teacher is the only pair where the raw student already reaches 55.3% (close to teacher's 63.7%), so most "signal" tokens carry tail-format detail rather than planning content; this is consistent with the mode-seeking caveat in §5.2 (teacher's argmax may exclude the right branch on long structured outputs). Both interpretations are post-hoc and would benefit from the suffix-only ablation queued in §7.

4.3 Stability and Efficiency

Stability. On three Qwen-math seeds (best-step avg@4 per seed), full-sequence has mean 56.78 ± 6.1 pp with the worst seed (50.45) *indistinguishable from* the no-distill baseline (50.95); pos-100 has mean 62.42 ± 2.9 with the worst seed at 60.40. The std estimate at n=3 has roughly 50% relative error, but pos-100's std is about 2.1× smaller and its worst seed beats full-seq's mean by 3.6 pp. A natural reviewer question is whether prefix-K is just early stopping in disguise: it is not—an early-stopped full-seq run (best-step selection across {50,

Table 4: Generality. Prefix-K best vs. full-sequence across model families, tasks, and methods. * indicates full-sequence collapse or post-training degradation. "HE" = HumanEval pass@1. **All cross-family rows are single-seed**; see Section 7. The 3-seed stability table is reported only for Qwen-math (Section 4.3).

Axis	Setting	Pos best	Full-seq	Note
Family (math)	Qwen 1.5B→1.7B	**66.65** (N=50)	62.35	—
	Gemma 2B→3-4B	**27.20** (N=100)	11.70*	full-seq below 13.45 baseline
	Llama 1B→3.1-8B (funcall)	**59.0** (N=150)	32.0*	full-seq degrades (funcall)
Task (Qwen)	Math (MATH-500 avg@4)	**66.65**	62.35	—
	Code (HumanEval)	**42.1**	40.2→26.8*	degrades
	Funcall (BFCL)	**61.3**	58.2	—
Method	LoRA math	**66.65**	62.35	—
	FullFT math	56.75	**58.20**	gap narrows
	FullFT code (MBPP+)	**47.4**	46.3	prefix wins
CoT thinking	Qwen3-1.7B→Qwen3-4B (avg@4)	**66.80** (N=100)	64.65	—

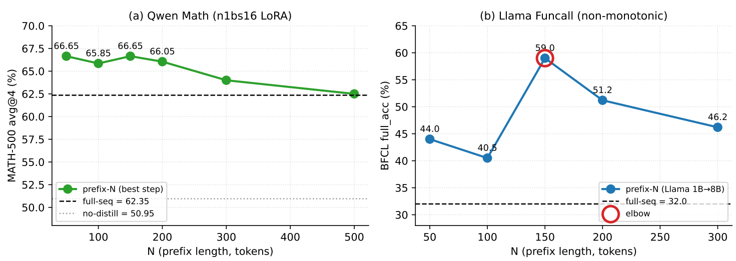

Figure 2: Position sweep with elbow. MATH-500 avg@4 vs. N. Plateau from $N = 50$ to $N \approx 200$; regression toward full-seq beyond.

100, 150, 200}, single seed) reaches 62.35 (Table 1), still 3.5–4.3 pp short of any prefix-K setting on that same seed. Prefix-K is stable where full-sequence is not, and dominates the early-stop full-seq baseline.

Efficiency. Per 200 training steps, full-sequence takes ~9.5 h and 38 GB peak memory; pos-100 takes ~1.0 h and 9 GB—a 9.5× wall-clock and 4× memory ratio. Most savings come from clamping generation length (`max_new_tokens` = N); the loss change is a one-line mask. No new hyperparameters.

4.4 Choosing N: a Single Elbow at 50–150

Figure 2 sweeps N on the canonical Qwen-math pair. The plateau extends from $N = 50$ to $N \approx 200$; performance regresses toward full-seq once N exceeds the useful prefix. A post-hoc rule of thumb fit on our three tasks: $N \approx 1.5\times$ the median length of the teacher's plan-plus-first-computation segment (estimated from a few teacher samples), giving $N \approx 50$ for code, ≈ 100 for math, ≈ 150 for function calling. We do not claim this generalizes. The two competing forces are coverage (each additional position adds some information) and noise (late positions encode execution and format rather than strategy).

Bridge to Part 3. Why does training a contiguous prefix beat training every token? Two regularities in our data point at distinct mechanisms: (i) late-position KL drops without late-position training, and (ii) the student exceeds the teacher exactly on the task (BFCL) where the teacher hedges most. Section 5 unpacks each.

5 Part 3: Why Position Works

Two complementary mechanisms account for the result. We frame Part 3 as *interpretation*—empirical regularities consistent with a specific story—rather than a proof. The headline result of Part 2 stands independently of this section being right.

5.1 Cascading Errors: Aligning the Planner Auto-Aligns the Executor

Evidence 1: tail KL drops without tail training. A model trained with pos-200tok is never updated on positions ≥ 200. Yet at inference time, KL on positions 200–400 drops to roughly the level achieved by full-sequence training (Table 5). The fix *propagates* downstream. The propagation is not unbounded: at positions 400–500, full-seq still beats pos-200tok (0.202 vs 0.289), so cascade alleviates but does not eliminate the need for late-position supervision when responses are very long.

Table 5: Tail KL drops without tail training. Per-position KL of the raw student vs. pos-200tok and full-seq students at step 200, on held-out trajectories. Bold rows are positions never touched by pos-200tok's loss.

Position range	Raw	Pos-200tok s200	Full-seq s200
0–50	2.064	1.015	1.036
50–100	0.759	0.322	0.312
100–150	0.495	0.228	0.236
150–200	0.387	0.248	0.236
200–300	0.382	**0.242**	0.273
300–400	0.331	**0.229**	0.308
400–500	0.317	0.289	0.202

Evidence 2: prefix swap. For each MATH-500 problem we generate a prefix of 100 tokens with model A and continue from token 100 onwards with model B (Table 6). Replacing only the prefix with the distilled student recovers ∼88% of the gain over the no-distill baseline ($(64.10 - 50.95)/(65.85 - 50.95) = 88.3\%$); replacing only the tail recovers ∼6%. The supervision lives in the prefix; the tail follows.

Table 6: Prefix swap, MATH-500 avg@4. "Pre-distill" = no-distill student; "Pos-100 student" = pos-100 distilled.

Prefix	Tail	avg@4
Pre-distill	Pre-distill	50.95
Pos-100 student	Pre-distill	**64.10**
Pre-distill	Pos-100 student	51.85
Pos-100 student	Pos-100 student	65.85

Intuition: conditional drift. If the student matches the teacher on the first k tokens, then for $t > k$ both distributions $p_\theta(y_t \mid y_{<t})$ and $q_\phi(y_t \mid y_{<t})$ are evaluated on the same context, and natural drift is bounded by the per-step on-policy disagreement (small in our experiments at most positions, see Table 5). If the prefix is wrong, every later conditional is evaluated on a *different* context, where on-policy disagreement compounds. Training the prefix is therefore not just locally cheap—it is globally load-bearing.

5.2 Mode-Seeking: How the Student Surpasses the Teacher

Disclaimer up front. What follows is one interpretation that is consistent with our data, not a proven mechanism. The hedging signal we report below is correlated with position, so we cannot fully separate "mode-seeking on a hedged distribution" from "the teacher's greedy decode is suboptimal at planning positions for unrelated reasons." The cleanest disambiguator would be a suffix-only ablation (§7); we present the evidence as suggestive, not conclusive.

Reverse-KL $D_{\mathrm{KL}}(p_\theta \| q_\phi)$ punishes the student for putting mass on tokens the teacher does not support, but does *not* punish concentrating mass on a single supported token. On a contiguous prefix, this commits the student to a single high-conviction trajectory

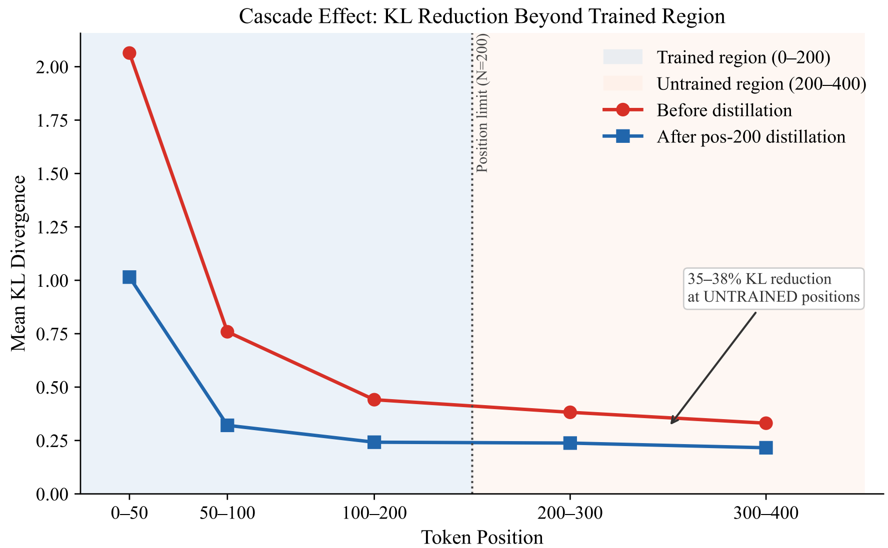

Figure 3: Cascade. Pos-200tok training (no tail loss) reduces inference-time KL on positions ≥ 200 to roughly the full-sequence level. The prefix carries the supervision and the tail follows.

mode. When the teacher hedges at a position—spreading mass across several plausible continuations—the student can still concentrate on one of them. The remainder of this section quantifies how much the teacher hedges at planning positions, which is precisely where reverse-KL has the most room to concentrate the student.

Evidence: student exceeds teacher on funcall. On BFCL (Table 7) the prefix-K student exceeds its teacher in two of three pairs. The teacher's greedy decoding is not the best policy on its own distribution; the mode-committed student is.

Table 7: BFCL function calling: student surpasses teacher in 2 of 3 pairs. Greedy full-accuracy, single-seed. "Student raw" = before distillation.

Pair	Teacher	Student raw	Pos-K student	Δ vs teacher
Qwen 1.5B→1.7B	54.0	2.7	**61.3** (N=100)	+7.3
Gemma 2B→3-4B	25.0	0.0	**30.9** (N=50)	+5.9
Llama 1B→3.1-8B	63.7	55.3	59.0 (N=150)	−4.7

Token-level evidence: the teacher hedges most where it matters. From `signal_analysis/range_summary.` (n=100 trajectories, Qwen math), Table 8 reports teacher entropy, teacher top-1 probability, and teacher–student argmax agreement by position range.

Table 8: The teacher hedges most at planning, agrees most at execution.

Position range	Teacher entropy	Teacher top-1 prob	T–S argmax agreement
0–50 (planning)	**0.259**	**0.909**	**0.785**
50–100	0.235	0.917	0.837
100–150	0.181	0.936	0.876
150–200	0.171	0.940	0.895
200–300	0.155	0.944	—

The teacher's entropy at positions 0–50 is 52% higher relative to positions 150–200, although both are quite peaked in absolute terms (top-1 probability ≥ 0.91 everywhere). The student-teacher argmax agreement gap is 11 pp wider at planning than at execution.

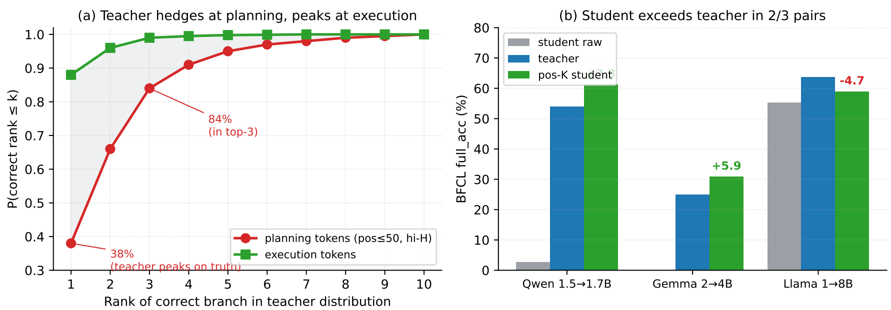

Figure 4: Student surpasses teacher. Left: teacher entropy and teacher–student argmax agreement by position range—the teacher hedges most at planning. Right: BFCL full-accuracy bars; pos-K students beat teachers on Qwen and Gemma pairs.

So the highest-stakes tokens are exactly the tokens where the teacher is least committed in *relative* terms, and where mode-seeking reverse-KL has the most room to concentrate the student on a single supported continuation. As flagged at the top of this subsection, this is consistent with mode-seeking but does not isolate it from a position-correlated confound.

6 Concurrent Work

A growing literature reweights or selects tokens for KD. Wang et al. [2025] report that the high-entropy 20% of tokens drive RL learning—a positional/entropy correlation finding rather than a fixed-budget selector. Huang et al. [2025] (SelectKD) and Xie et al. [2026] (AdaKD) use *soft confidence reweighting*: every token retains gradient, with weights set by teacher confidence or KL. Kim and Baek [2026] (TSDKD) anneals an entropy threshold over training rather than fixing it. Li et al. [2025] document a planning-then-anchor rhythm in attention without proposing a selector. Chen et al. [2025] use prefix truncation as an off-policy efficiency heuristic without comparing alternatives. Our paper contributes orthogonal evidence to all of these: a head-to-head comparison at *fixed budget* K=100, with negative controls (middle/last) showing late-token training is harmful, plus the cascade and mode-seeking analyses.

Scope of comparison. We test entropy and confidence as token *selectors* (hard top-K); the cited methods deploy them as *soft reweightings* of the full-sequence loss. The two are not interchangeable. Our claim is therefore narrower than "position beats every per-token-magnitude method": it is that, when reduced to fixed-K hard selection, position dominates magnitude. Whether soft entropy/confidence reweightings of the full-sequence loss can match or exceed prefix-K remains an open empirical question, and is the most natural follow-up to this work.

Sequence-level KD [Kim and Rush, 2016, Gu et al., 2024] uses one loss per sequence and does not ablate which tokens carry the gradient.

7 Limitations

Scale. Largest student is 8B; the elbow at 70B+ is untested. **Long-form.** Responses are <1k tokens; multi-thousand-token CoT is untested. **Suffix-only ablation queued.** A clean test of "planner-tail asymmetry: position or content type?" is not yet finished. **Mode-seeking assumes coverage.** If the teacher's top-K excludes the correct branch, mode-seeking commits to the wrong mode—observed on Llama BFCL (−4.7 pp vs. teacher). **Single-seed cross-family.** The 3-seed stability table is Qwen-math only. **Falsifier.** A budget-unconstrained format-mask full-sequence loss matching prefix-K would reduce "position" to "content type"; our K=100 format-mask ties full-sequence (62.05 vs. 62.35) but does *not* reach prefix-K (66.65). We claim a robust empirical regime (OPD reverse-KL at small-to-mid scale) where position uniquely exceeds full-sequence, with an interpretation consistent with all our evidence—not that position is a universal prior or that cascade is proven.

8 Conclusion

At fixed budget K=100, top-KL underperforms full-sequence by 3.75 pp, entropy variants tie, and only the contiguous prefix *exceeds* full-sequence—by 3.5–4.3 pp on math, in 7 of 9 cells across families, methods, and tasks; late-token training is actively harmful. Prefix-K is one-line, $\sim$10$\times$ faster, $\sim$4$\times$ smaller in memory, has a single elbow at $N \approx$ 50–150, and lets students surpass their teachers on BFCL in 2 of 3 pairs by 5.9–7.3 pp. Cascade and mode-seeking together account for the result. The narrower practical takeaway: under fixed-budget hard selection, magnitude indicators do not exceed full-sequence; a contiguous prefix does, at lower cost. Two questions remain open: whether soft reweightings can match prefix-K at the same budget, and whether the prefix advantage is content-driven (LaTeX/format-token noise concentrated in the tail) or position-driven—the format-masked full-sequence ablation in §7 is the disambiguator.

References

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, 2024.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Wei-Rui Chen, Vignesh Kothapalli, Ata Fatahibaarzi, Hejian Sang, Shao Tang, Qingquan Song, Zhipeng Wang, and Muhammad Abdul-Mageed. Distilling the essence: Efficient reasoning distillation via sequence truncation. *arXiv preprint arXiv:2512.21002*, 2025.

DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. In *International Conference on Learning Representations*, 2024.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems*, 2021.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Haiduo Huang, Jiangcheng Song, Yadong Zhang, and Pengju Ren. SelecTKD: Selective token-weighted knowledge distillation for LLMs. *arXiv preprint arXiv:2510.24021*, 2025.

Minsang Kim and Seung Jun Baek. Explain in your own words: Improving reasoning via token-selective dual knowledge distillation. *arXiv preprint arXiv:2603.13260*, 2026.

Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1317–1327, 2016.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*, 2023.

Yang Li, Zhichen Dong, Yuhan Sun, Weixun Wang, Shaopan Xiong, Yijia Luo, Jiashun Liu, Han Lu, Jiamang Wang, Wenbo Su, Bo Zheng, and Junchi Yan. Attention illuminates LLM reasoning: The preplan-and-anchor rhythm enables fine-grained policy optimization. *arXiv preprint arXiv:2510.13554*, 2025.

Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

Qwen Team. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024.

Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning. In *Advances in Neural Information Processing Systems*, 2025.

Xurong Xie, Zhucun Xue, Jiafu Wu, Jian Li, Yabiao Wang, Xiaobin Hu, Yong Liu, and Jiangning Zhang. LLM-oriented token-adaptive knowledge distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026.

A Coding Results

Table 9: Coding (HumanEval/MBPP pass@1, Qwen LoRA, baseline HE=32.9, HE+=29.3).

Method	Best HE	Best HE+	Best MBPP	Best MBPP+	Best step
LoRA pos-50	**42.1**	**36.6**	46.6	41.5	350
LoRA pos-100	**42.1**	36.0	49.2	44.4	150
LoRA pos-150	41.5	**37.8**	49.5	44.7	250
LoRA fullseq	40.2	35.4	**52.6**	**46.3**	50 (then degrades)
FullFT pos-250	**37.8**	**31.7**	**54.2**	**47.4**	350
FullFT fullseq	36.6	31.7	53.2	46.0	150

B Surprise Quartiles in the Head

Table 10: Surprise-quartile distribution among the first N tokens (74,549 tokens, Qwen-math). Q1 = top 25% surprise globally.

Head N	n	Q1	Q2	Q3	Q4	Q1 enrichment
25	2,500	**70.0%**	14.0%	8.6%	7.4%	2.8×
50	5,000	**61.0%**	14.7%	11.2%	13.1%	2.4×
100	9,991	53.2%	17.1%	12.4%	17.3%	2.1×
200	19,856	42.6%	20.0%	14.3%	23.1%	1.7×
500	44,553	31.8%	22.3%	18.8%	27.1%	1.3×

C Token Classification Details

Token classification was performed on ∼2M tokens from 10K trajectories using rule-based categorization over the Qwen2.5-Math-1.5B tokenizer. Categories: **planning** (sentence-initial strategy words: "To", "Let", "First", "Therefore", "Thus", "Since", "Given"); **structural** (`**`, `:`, newlines, `$`, parentheses, brackets); **math_latex** (`\(`, `\[`, `\)`, `\]`, `\frac`, etc.); **math_number** (digits 0–9); **math_operator** ($=, +, -, \times, /$); **continuation** (all remaining).

Position 0 has mean KL = 7.71 (10× the overall mean of 1.01). The dominant token is "To" (75.1% of trajectories, KL = 7.32). The teacher prefers more diverse openings: "First" (KL = 14.0), "Step" (KL = 14.6), "Solution" (KL = 21.9). Position 0 alone accounts for 3.8% of total KL signal across the entire trajectory.

D Reproducibility

The complete training script (`on_policy_distill_positional.py`) supports `-position_limit N`, `-max_new_tokens N`, `-full_finetune`, `-token_select_mode {prefix, top_kl, top_ent_student, top_ent_teacher, format_mask, random, middle, last}`. All n1bs16 LoRA experiments use $r = 32$, $\alpha = 64$, lr 5×10^{-5}, temperature 0.7, 200 steps, 3200 problems, $n = 1$, batch size 16. MATH-500 evaluation uses vLLM [Kwon et al., 2023] with `max_model_len`=4096 and $n = 4$ samples at $T = 0.7$.